

Towards Improving Enterprise Performance with Service Level Agreements

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Abstract— The BPM-SOA combination is being advocated as the best approach for enterprises to reach the desired business agility and responsiveness to changing business requirements as it brings the service-based business processes employing loosely-coupled services. However, in order to have a certain business, the behavior and non-functional properties of these services such as response time, throughput and availability should be guaranteed and this can be specified by Service Level Agreements (SLAs). In this regard, our work expresses SLAs for the enterprise domain in a machine understandable format. we propose an ontology-based representation model of SLA and utilize the business services performance requirements specified as Key Indicators (KPIs and KQIs) to define SLA parameters. The purpose is to make it possible to monitor business processes based on SLA in order to assure compliance with business requirements and targeted objectives. In general, this model can help automate the process of SLA representation, monitoring and taking actions in case of violations in SOA and business process domains. At the end, an SLA monitoring prototype system is proposed to show how our model can be deployed.

Keywords-Service Level Agreement (SLA); Ontology; SWRL; Key Performance Indicator (KPI); Key Quality Indicator (KQI)

I. INTRODUCTION

In the modern business environments, enterprises are paying more attention to the management of their business processes in order to promote their business efficiency. On the other hand, the service oriented architecture (SOA) and its most common implementation, services, enable the enterprises to increase their agility in the face of change, improve their operating efficiency, and greatly reduce the cost of doing business. Therefore the BPM-SOA combination is being advocated as the best approach for enterprises to reach the desired business agility and responsiveness to changing business requirements. In the SOA-based enterprises, business processes and services are coupled with each other to fulfill dynamic business demands. In this vision, a business process can be seen as a set of services and can be choreographed with other business processes or services, even between different enterprises and organizations. When designing such a service-based business process employing loosely coupled services, it is important to be ensured that the process is executed as expected. For this purpose, the service provider has to specify both non-

functional properties as well as functional, in particular the technical quality of service characteristics such as response time, throughput and availability.

In order to define expected service behavior and quality of the interaction with regard to the non-functional properties of service, service provider and service consumer create a contract called Service Level Agreements. SLAs are an important tool for guiding and guaranteeing the quality of the execution of service-based business processes and the achievement of business goals because from service perspective, they can describe each involved actor in the business process model by both functional and trustworthy capabilities so that qualified services could be aggregated to fulfill the business goal.

However, SLA definition is not straightforward from the business goals of the organization running the processes [1]. This research uses KIs (KPIs and KQIs) concept as a good solution to identify the essential metrics and parameters of SLA in order to map business goals to business services requirements. The KI is a key instrument to evaluate the performance of business services and detect the state of current and completed processes [2]. In this work, we use KIs which TMForum is proposed as business services key indicators in order to map to SLA parameters. Also, we develop an ontology to represent SLAs and KIs in a machine understandable format in an enterprises environment. The need for automatically representation SLAs is driven by future delivery of complex new generation of services, and by the need to reconfigure services quickly in response to change in requirements. Then we utilize Semantic Web Rule Language (SWRL) [3] to infer hidden relationships among KIs as SLA parameters. In this way, the compliance of SLAs can be checked by comparing the KIs values measured over business processes with those agreed in SLAs and also it would be possible to report violations.

The remainder of the paper is organized as follows: section 2 describes the basic concepts in order to facilitate the understanding of our approach. Section 3 presents an overview of related work in this area. The proposed model is presented in section 4. Section 5 presents a prototype system that explains how our model is used for monitoring SLA. At last, conclusions and an outlook to our future work are discussed in section 6.

II. BACKGROUND

A dynamic view of an enterprise is reflected in its business processes [4]. A business process is the combination of a set of activities within an enterprise with a structure describing their logical order and dependence whose objective is to produce a desired result [5]. In today's dynamic business environment, enterprises need to respond more effectively and quickly to changes and opportunities in order to remain competitive in the global markets. For this reason, business processes need to be modeled with e-service applications which bring unassociated and loosely coupled units of functionality that are coupled with each other to fulfill dynamic business demands. Since in our study, we assume that business process activities are associated with Web services [6] and can be implemented using, for example, the business process execution language for web service (BPEL4WS) [7].

From a business perspective, a generalized statement of business goals relevant to the scope of the project is decomposed into sub goals that must be met in order for the higher-level goals to be met. This hierarchical decomposition of the goals leads to identify the services that will help in fulfilling the sub goals. It is also necessary to identify key indicators in order to provide an objective basis for evaluating the degree to which the goal has been achieved. Key indicators and target values identified during the process execution are used to measure, monitor, and quantify the success of the SOA solution in fulfilling business needs [4].

In this regard, we use the key indicators that are presented in the SLA Management Handbook [8]. This handbook is a publication of the Tele Management Forum (TMF) and part of the New Generation Operation Support Software (NGOSS) project. Although the contents of the Handbook are basically aligned to business in the telecommunication industry, they are generally also applicable to a border scope of IT-dependent organizations, because most of the presented concepts are not restricted to telecoms industries specifics [9]. The Handbook utilized KPIs and KQIs for managing service quality. KPIs are quantifiable measurements that reflect the critical successful or unsuccessful factors of a particular service. KPIs represent the performance; thus, they cannot completely represent end-to-end service quality, therefore the TMF proposed KQIs, which are indicators that provide measurements of a specific aspect of the performance of the product, product components (services) or service elements, and represent their data from a number of sources including the KPIs. The Handbook also defined a hierarchy among KPIs, KQIs, and SLAs as shown in "Fig. 1". SLAs in which provider and consumer define the expected service behavior and its quality, can be defined in terms of services KQIs.

Service KQIs use service KPIs as metrics for reporting the performance of their services (Target values for KPIs must be reached in a certain period). An example of the relationship between KPIs and KQIs is shown in "Table 1"

in which the business service is Video Conference. In our work the SLA concept is oriented to the service relationship between service consumer and service provider in which a set of metrics could be used for describing and measuring levels of quality of service. This is in conformity with the definition contained in the SLA Management Handbook of the TeleManagement Forum.

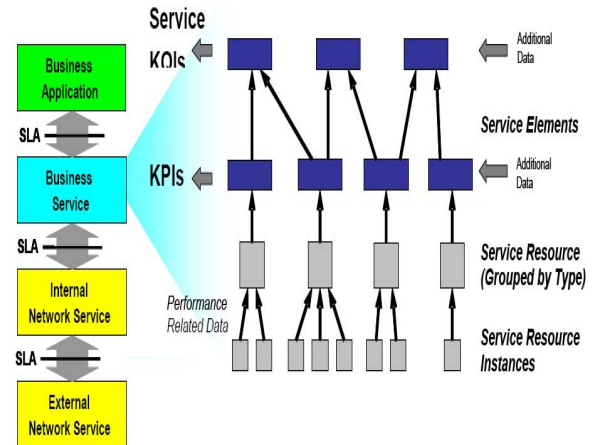


Figure 1. Relationship between SLA, KQI, and KPI [8].

Metrics are used to measure and manage performance compliance to SLA commitments, and are the heart of a successful agreement as the critical long term success factors [10]. The categorization of SLA contents with a particular focus on SLA metrics facilitates design decisions and helps to identify responsibilities for critical IT processes in disruption management during the execution of SLAs [11].

In this research, presented KPIs and KQIs are mapped to these metrics.

III. RELATED WORK

A close approach to ours is presented in [12] for management of SLA-aware service compositions based on process performance requirements specified as KPIs. The proposal is to use KPIs in order to help define the service level objectives specified in the SLAs that must be met by provider services and IT Infrastructure. According to their proposal, the performance requirements on the process are gathered and mapped to dependent service level objectives of partner service. Then optimal services are selected and at last, during process execution, SLAs are monitored. But indeed the authors just present an outline of methodology explaining the steps to do but they do not determine how to perform it.

Table1. KQI and KPI Relationship

Service	Service KQI	Service KPI
Video Conference	Availability	MTBF, MTBR
		Loss of Service
	Speech/Visual Quality	MOS
		Loss, Jitter, Delay
		CustomerSatisfaction
	Response Time	Response Time
	Round Trip Delay	OWD, RTT
	Delay	OWD, RTT
	Confidentiality	PhysicalAccessViolations
	Non-repudiation	PhysicalAccessViolations
	Interoperability	InteroperabilityComplaints
	Connect Time	Connect Time

A model towards web SLAs specification and monitoring through Web Services technology is WSLA [13]. The Web Service Level Agreement language (WSLA) is defined as an XML Schema and covers the definition of the involved parties, the services guarantees and the service description. It comprises the following major parts: *parties*, *service definition* and *obligations*. The WSLA can be applied for inter-domains management in business-oriented scenarios for the processes and general applications. Considering this aspect, the WSLA approach has several similarities with our proposal. The KPI and KQI concepts are not used, but according to the WSLA, a service has SLA parameters, defining metrics which are expressed or evaluated in terms of cascaded functions. The possibility of creating metrics using the functions is not explicitly supported by the existing version.

Another work that most closely relates to ours ontology-based SLA definition is described in [14, 15]. It proposes an ontology-based model of SLAs that is used to automatically generate semantic-enabled QoS obligations monitors. The difference is that we have used KIs for defining SLA terms and moreover, our work expresses SLAs for the enterprise domain and focuses on business process management in an SOA-based enterprise. We have used SWRL reasoning rules for representing KIs dependencies and automatically evaluating them against the SLA parameters and inferring the occurrence of SLA violations. With this way, it is possible to assure the traceability between KPIs defined over business services and their target values established in SLAs and fulfill the business goals of the organization running the processes.

There are several models (we do not explain here due to lack of space) like [16] that have used XML elements to define KPI and SLA models. These methods have advantages but in fact, using fixed rules to define SLA

violations is less robust than using ontologies for this purpose. The ontologies offer inference possibilities and semantic interpretations, especially when they are implemented using the OWL language.

IV. THE ONTOLOGICAL SLA MODEL

In this work, it is assumed that the service provider offers the services with desired functionality, so we focus on services non-functional requirements. The correct management of such requirements directly impacts the success of organizations participating in e-commerce. To achieve this, a number of research contexts need to be explored. First, mechanisms are necessary to synthesize business services KIs on the SLA metrics of the web services that compose the business process. Then, a good theoretical SLA model is necessary to formally specify and represent SLA. Finally some approaches need to be developed to monitor and manage SLAs.

In this regard we model SLA metrics based on performance information of the services that compose the business process. Our approach relies on the use of ontologies to represent the SLA model. Ontology-based approaches have been suggested as a solution for semantically describing SLA specifications in a machine understandable format. Then an autonomic SLA monitoring system can be developed using SWRL rules and the Jess rule engine. The purpose of this system is detecting SLA violations. In this way, there is a possibility to assure the traceability between KPIs defined over business services and their target values established in SLAs.

In this section, we illustrate the ontological model for representing SLA based on KIs. As mentioned before, OWL is chosen to express the ontology and its elements are used for describing the entities and relationships related to SLAs. The relation between KPIs, KQIs and SLA parameters are defined through SWRL rules. We adopt UML class diagram for representing the ontological SLA model which is a semantic extension of the WSLA specification.

In general, an SLA is a contract between a service consumer and a service provider over a specific service. It contains a description of the service functionality, definition of service level parameters and statement of responsibilities of both parties. Also, it includes penalties for service level violations to pay by the service provider [9]. Therefore our ontology-based SLA should contain all of these entities, as shown in “Fig. 2”. In this model, *SLA* is the main class and aggregates three main subclasses: *Parties*, *Services* and *Obligation*. *Parties* class describes consumer and provider involved in an SLA. It aggregates other classes used to store information such as name, phone number, address, email and other related data. *Services* class represents the information about the offered services like the service level that is specified through service parameters and their corresponding KPIs. *Obligation* class represents the

conditions that must be respected by the provider with respect to the offered service. These conditions are expressed in terms of KQIs, which are used for defining the terms under which the offered service will be monitored and evaluated. Preferentially, KQIs are expressed as a function of KPIs. 'Availability' is a KQI example that can be defined in terms of 'MTBR' and 'MTBF' KPIs for Helpdesk service. *Obligation* class also defines the penalties to be applied when the expressed conditions are offended. We define how a KQI is calculated in terms of the associated KPIs using SWRL rules.

SWRL rules are chosen to express SLA obligations and infer relationships among KIs, calculating KQIs from KPIs and detect SLA violations. The represented ontology-based model involves relationships between KIs and SLA which are not one-way, and some of them are transitive. These unspecified relationships could be found by using SWRL rules. In the following, several instances and corresponding SWRL rules are explained. The following SWRL rules complement the one-way *influence* relationship between two KPIs by adding the other relationship.

- $SLAParameter(?x) \wedge SLAParameter(?y) \wedge influence(?x, ?y) \rightarrow isInfluencedBy(?y, ?x).$
- $SLAParameter(?x) \wedge SLAParameter(?y) \wedge isInfluencedBy(?x, ?y) \rightarrow influenced(?y, ?x).$

Also, the following SWRL rules complement the one-way *hasSLAParameter* relationship between a KQI of an SLA and a KPI by adding the other relationship.

- $KeyQualityIndicatorSet(?x) \wedge SLAParameter(?y) \wedge hasSLAParameter(?x, ?y) \rightarrow isSLAParameterOf(?y, ?x).$
- $SLAParameter(?x) \wedge KeyQualityIndicatorSet(?y) \wedge isSLAParameterOf(?x, ?y) \rightarrow hasSLAParameter(?y, ?x).$

Similarly, more SWRL rules could be easily defined and added in order to calculate more KPIs. Also, by using SWRL rules, it can be checked whether an SLA is satisfied or not. *SLO*, which is a main element of the SLA model, comprises the *Threshold*, *CompareFunction* and *KeyQualityIndicatorSet* subclasses. The SLA violations can be detected by comparing *Threshold* and *KeyQualityIndicatorSet* value using the *CompareFunction*, as shown in the example below:

- $SLA(?sla) \wedge hasObligation(?sla, ?obl) \wedge hasKeyQualityIndicatorSet(?obl, ?indicator) \wedge hasThreshold(?sla, ?threshold) \wedge hasCompareFunction(?obl, ?operator) \wedge hasValue(?indicator, ?value) \wedge swrlb : equal(?operator, "gt") \wedge swrlb : lessThan(?value, ?threshold) \rightarrow isViolated(?sla, "true"),$
- $SLA(?sla) \wedge hasObligation(?sla, ?obl) \wedge hasKeyQualityIndicatorSet(?obl, ?indicator) \wedge hasThreshold(?sla, ?threshold) \wedge hasCompareFunction$

$(?obl, ?operator) \wedge hasValue(?indicator, ?value) \wedge swrlb : equal(?operator, "gt") \wedge swrlb : greaterThanOrEqual(?value, ?threshold) \rightarrow isViolated(?sla, "false").$

More SWRL rules can be defined using a different *CompareFunction* in order to detect SLA violations.

V. SLA MANAGEMENT SYSTEM

Based on our model, we have developed an ontology for representing SLA specification and presented a Web Ontology Language (OWL)/Semantic Web Rule Language (SWRL) based Knowledge Base that can be used in an autonomous SLA management system. In our study, Protégé, a free open-source ontology editor [17] with other related plug-ins like SWRL tab are employed. In order to implement our ontology, we built *SLA* OWL, and then, the SWRL rules have been added for inferring hidden relationships among KIs or between an SLA and a KI, calculating KQIs from KPIs, and detecting SLA violations. In order to implement this inferring, we utilized Drools rule engine [18] which supports the reasoning using knowledge that is supplied in the form of declarative rules. The SWRL Drools tab is a plug-in to the SWRL tab in Protégé-OWL.

The proposed system only includes those concepts of standards which are essential to model functional dependencies between process activities, service operations and service implementing component functions within a SOA. The service operations can be invoked by process activities, which are linked in the context of a business process. In this work, only invoking activities with a request-response interaction model are focused.

For modeling metric dependencies between services and processes, we focus on metrics which can be measured or calculated at runtime. Examples for such metrics are response time and availability [19].

A business process is implemented using BPEL and is deployed on a *BPEL Engine*. The engine manages execution of business processes: coordinates the execution of activities, providing input to an activity, binding and execution of a web service associated with the activity, routing the request to appropriate activity depending upon the output obtained. The BPEL process invokes the service operation of the SOAP-based web service described by WSDL. The binding between the process and the service could be done statically or dynamically. (A *Web service Implementation* exposes its functionality via a service interface described by WSDL). *Monitor* is responsible for monitoring and measuring the SLA parameters based on *BPEL Engine* monitoring component. *Expert System* allows assertion of facts in the knowledge base depending upon the rules selected. *Knowledge Base* is a repository of facts about SLA related parameters. *Rule Repository* collects rules used to specify KIs requirements.

In order to show the relevance of this proposal in practice, we consider an exemplary service provider that provides Video Conference business service for service

consumer. The consumer and the service provider agreed on the SLA that parts of its obligation information are given in “Table 2”. Three types of *KeyQualityIndicatorSet* including ‘Availability’, ‘Response Time’ and ‘Speech/Visual Quality’ are considered and each of them has their own *Parties*, *CompareFunction* and *Threshold*.

The *KeyQualityIndicatorSet* vlues are calculated based on *KeyPerformanceIndicatorSet* using *KQIFunction* at execution time. Then, the defined SWRL rules (described in Section IV) are executed using the Jess rule engine to figure out whether the related SLA has been violated or not. The results show that quality indicators of ‘Availability’ and ‘Speech/Visual Quality’ are satisfied but ‘Response Time’ is violated.

VI. CONCLUSION AND FUTURE WORK

The increasing role of SLAs in enterprise systems is considerable. In B2B environments, the SLA mechanism helps service provider and consumer manage the risks and expectations, and establish trusted business relationships. Effective SLAs are very important to guarantee business continuity, customer satisfaction and trust. In general, being able to characterize business process based on SLA parameters has some advantages for organizations. First, it allows for more efficient translation of organizations vision into their business processes, since those can be designed according to service specifications. Second, it allows the selection and execution of web services based on business process requirements, to better fulfill customer expectations. Third, it makes possible the monitoring of business processes based on SLA. Business processes must be rigorously and constantly monitored throughout their life cycles to assure compliance with initial KIs requirements and targeted objectives. In order to achieve these purposes, we introduced a model to manage service-based business processes with SLAs that guarantee a certain quality of performance. In this regard, we have investigated business services KIs hierarchy based on [8] and proposed the ontology-based SLAs and SWRL rules that are used for inferring hidden relationships among KIs and SLAs.

There are a number of restrictions and open areas in this work that is explained in following. We considered and classified almost every KPIs based on [8], however, it needs to be verified whether it reflects the actual consumer and service provider’s criteria. Moreover, in order to cover all aspects of the KPI hierarchy, many other SWRL rules have to be integrated with our ontology. Another restriction is about violations. Neither party wants the SLA to be violated; consumers want a high level of service for their key business processes, not a payment for SLA violation, which will never compensate for the loss of business. Similarly the provider does not want to suffer the loss of market trust and credibility which may affect many more accounts than that affected by the SLA violation. Therefore the SLA must be in place to foster a cooperative business approach to common [20]. Hence, forecasting SLA violations is more appropriate

than just detecting them and it is an important part of our future work.

In spite of the above mentioned restrictions, our ontology-based SLA has some advantages as it is very easy to extend due to its use of ontologies. However, one of our major research aims for the future work is finding a suitable way to forecast SLA violations in SOA-based enterprise environment.

REFERENCES

- [1] G. Frankova, M. Séguranb, F. Gilcher, S. Trabelsi, J. Dorflinger and M. Aiello, “Deriving business processes with service level agreements from early requirements,” *Journal of Systems and Software*, Elsevier, vol. 84, 2011, pp. 1351–1363, doi:10.1016/j.jss.2011.03.077.
- [2] E. Toktar, G. Pujolle, E. Jamhour, M. Penna and M. Fonseca, “An XML model for SLA definition with key indicators,” *IP Operations and Management*, Springer, vol. 4786, 2007, pp. 196–199.
- [3] I. Horrocks, P. F. Patel-Schneider, H. Boley, S. Tabet, B. Groszof and M. Dean, “SWRL: A Semantic Web Rule Language combining OWL and RULEML,” W3C member submission, W3C, May 2004.
- [4] A. Arsanjani, S. Ghosh, A. Allam, T. Abdollah, S. Ganapathy, and K. Holley, “SOMA: A method for developing service-oriented solutions,” *IBM Systems Journal*, vol. 47(3), 2008, pp. 377–396.
- [5] R. S. Aguilar-Saven, “Business process modelling: Review and framework,” *Elsevier*, vol. 90, 2004, pp. 129–149, doi:10.1016/S0925-5273(03)00102-6.
- [6] G. Shegalov and M. Gillmann, “XML-enabled workflow management for e-services across heterogeneous platforms,” *The VLDB Journal*, 2001, vol. 10(1), pp. 91–103.
- [7] Z. Li, W. Sun, Z.B. Jiang, and X. Zhang, “BPEL4WS Unit Testing: Framework and Implementation,” *Proc. IEEE international Conference on Web services (ICWS’05)*, July 2005, pp. 103–110.
- [8] The TeleManagement Forum and the Open Group, “SLA Management Handbook,” vol. 4: Enterprise Perspective, 2004.
- [9] T. Schaaf, “Frameworks for Business-driven Service Level Management: A Criteria-based Comparison of ITIL and NGOSS” in *Proceedings of Second IEEE/IFIP International Workshop on Business-Driven IT Management (BDIM 2007)*, 2007.
- [10] I. Hayes, “Metrics for IT Outsourcing Service Level Agreements,” Available at www.clarity-consulting.com/metrics/article.htm, 2004.
- [11] A. Paschke and E. Schnappinger-Gerull, “A Categorization Scheme for SLA Metrics” in *Multi-Conference Information Systems (MKWI06)*, 2006. Passau, Germany.
- [12] B. Wetzstein, D. Karastoyanova and F. Leymann, “Towards management of sla-aware business processes based on key performance indicators,” In *9th Workshop on Business Process Modeling, Development and Support (BPMDS’08)*, France, 2008.
- [13] H.Ludwig, A. Keller, A. Dan, R. Franck and R.P. King, “Web Service Level Agreement (WSLA) Language Specification” IBM Corporation, July 2002.
- [14] K. Fakhfakh, S. Tazi, K. Drira, T. Chaari and M. Jmaiel, “Semantic enabled framework for sla monitoring” *International Journal on Advances in Software*, 2009, vol. 2(1), pp. 36–46.
- [15] K. Fakhfakh, T. Chaari, S. Tazi, M. Jmaiel and K. Drira, “ODACE SLA: Ontology Driven Approach for Automatic Establishment of Service Level Agreements” *International Journal of Systems and Service-Oriented Engineering*, 2010, vol. 1(3), pp. 1–20.
- [16] E.Toktar, G. Pujolle, E. Jamhour, M. Penna, and M. Fonseca, “An XML model for SLA definition with key indicators” *IP Operations and Management*, Springer, 2007, vol. 4786, pp. 196–199.
- [17] The protege ontology editor and knowledge acquisition system, Available: <http://protege.stanford.edu/>

[18] The Rule Engine for the Java™ Platform, Available: <http://www.jessrules.com>

[19] S. Kalepu, S. Krishnaswamy and S. W. Loke, "Verity: a qos metric for selecting web services and providers," in 4th International Conference on Web Information Systems Engineering Workshops, Dec. 2003.

[20] B. Mitchell and P. McKee, "SLAs A Key Commercial Tool," In P. Cunningham and M. Cunningham (eds.), Exploiting the Knowledge Economy- Issues, Applications, Case Studies, Proc. eChallenges e-2006 Conference, IOS Press, Oct. 2006, vol. 3.

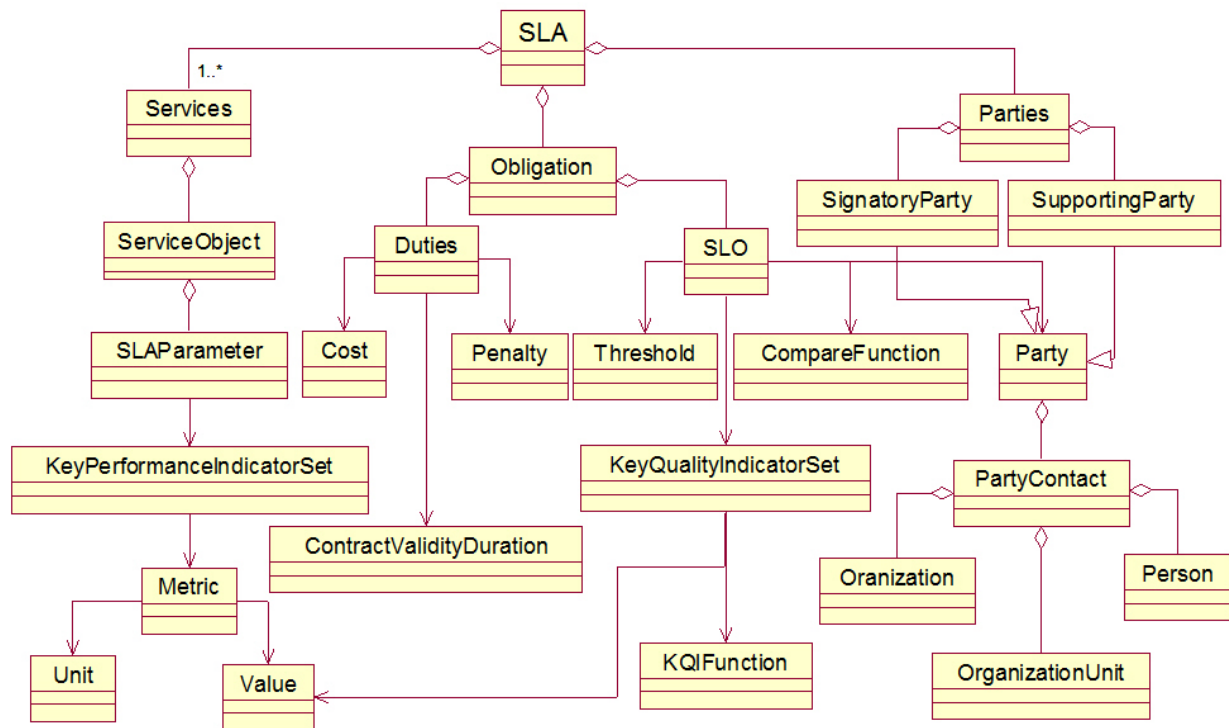


Figure 2. SLA Class Diagram

Table2. SLA Management System Test

Service	KeyQualityIndicatorSet	Threshold	CompareFunction	The values obtained	Result
Video Conference	Availability (MTBF, MTBR)	98%	greater than or equal	98%	satisfied
	Speech/Visual Quality (MOS Loss, Jitter, Delay CustomerSatisfacation)	95%	greater than	96%	satisfied
	Average Response Time	1min	less than	2min	violated